Lecture 2: The Graphical Method

Week 2: The Graphical Method

ISE 3200 - Linear Optimization

Feasible regions

- Plot each constraint as a half-plane
- The feasible region is their intersection

Corner-point theorem

- If an optimum exists, one is at a corner point
- Evaluate z at every corner, including line intersections

Special cases

- Unbounded: the region extends in an improving direction
- Multiple optima: objective parallel to a binding edge
- Infeasible: empty region